ALEXANDER MEJÍA OSPINA

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PROFESSIONAL SUMMARY

Seventh-semester Software Engineering student who builds and ships complete projects: a deployed Spring Boot REST API with tests, a data pipeline over 1.18 million public health cases with Power BI dashboards, a Flutter app, and a React website in production. Available now for a full-time software or data internship.

EDUCATION

Universidad Cooperativa de Colombia — Medellín, Colombia
B.S. in Software Engineering — 7th semester (expected graduation: 2027) — 2023 – Present
Relevant Coursework: Data Analytics, Software Testing, Software Implementation and Maintenance, Numerical Methods.
Scholarships: Recipient of the Sapiencia – Participatory Budget scholarship (Medellín).

Colegio Parroquial San Buenaventura — Medellín, Colombia
High School Diploma (Academic Track) — 2012 – 2022

TECHNICAL SKILLS

Languages: Java, Python, Dart, JavaScript, C++, SQL, HTML, CSS
Backend: Spring Boot, Spring Data JPA, REST APIs, Swagger/OpenAPI, FastAPI, Flask, Maven
Data Analysis: SQL (CTEs, window functions), Python, Pandas, Power BI, Looker Studio, Jupyter, Google Colab, regression
Web and Mobile: HTML, CSS, Bootstrap, JavaScript, React, Flutter, Riverpod, MVVM architecture
Testing and DevOps: JUnit, Mockito, pytest, Postman, Docker, GitHub Actions (CI), Git, GitHub
Databases and Cloud: MySQL, SQLite, Firebase, Google Cloud Platform, Render
Methodologies: Scrum, software testing (test lead on an academic project) | **Spoken Languages:** Spanish (native), English (intermediate, B1)

PROJECTS

Antioquia Public Health Observatory (live · code) — *Personal project* — 2026
- Built a Python and Pandas ETL pipeline that downloads 1.18 million cases reported to Colombia's SIVIGILA surveillance system (2007–2022) from open data, cleans and validates them, and loads them into a MySQL star schema.
- Wrote SQL queries with CTEs and window functions for trends, municipal rankings, and endemic channels; published the findings on a website, a FastAPI API, and Power BI (DAX, rate per 100,000) and Looker Studio dashboards.
- Automated pytest tests and monthly data refreshes with GitHub Actions.

CineX — Cinema Management System (live · code) — *v1.0 team, v2.0 solo* — 2025 – 2026
- v1.0 (team): set up the repository, integrated the team's work, and reviewed pull requests for a Java 21 Spring Boot REST API and its web admin panel.
- v2.0 (solo): moved movies, rooms, showtimes, and bookings to MySQL with Spring Data JPA, adding business rules (room capacity, no overlapping showtimes), validation, and JSON errors documented with Swagger.
- Added 19 JUnit and Mockito tests, a 26-request Postman collection, Docker Compose, and GitHub Actions CI; the API is deployed on Render with MySQL on TiDB Cloud.

Numi — Offline Educational App with AI Assistant — *Team project* — 2026
- Flutter app for elementary students in rural Medellín that works without internet, with 15 activities across 5 subjects; MVVM architecture with Riverpod and progress synced to a FastAPI backend.
- My role: designed the app architecture, built the logic and screens for Natural Sciences, the offline RAG for the AI assistant (on-device Gemma 2B with a SQLite knowledge base), and the responsive design.
- Admin and teacher web panels (Flask, MySQL) with Luis Miguel Muñoz, connected to the app and covered by 110 tests.

Casa Rendón — Website (casarendon.com) — *Team project, in production* — 2026
- Designed and built the website of an artisanal coffee and wine brand: React components for the catalog, gallery, and FAQ, and a visit-booking form that sends requests via WhatsApp.

Digital Dental Records — Capstone Project — *Team proposal, UCC* — 2026 – 2027
- Web app so Oral Health Assistant students record each patient visit and their teacher approves it without paper: 17 requirements, 9 use cases, a 5-alternative decision matrix, and a modular monolith design; built with Scrum in 2027.

Data Analysis with Python — *Data Analytics course* — 2026
- 23 Google Colab labs: EDA, statistics, regression (R^2 0.69), k-means, time series, and a neural network with 97.6% accuracy.

ADDITIONAL TRAINING

Java and JavaScript Certificate Programs — Politécnico de Colombia · **C++ Essentials 1** — Cisco Networking Academy — 2025
Cloud: Google Cloud — 4 Google Skills badges · AWS Academy Cloud Foundations (in progress) — 2026

REFERENCES

Available upon request.